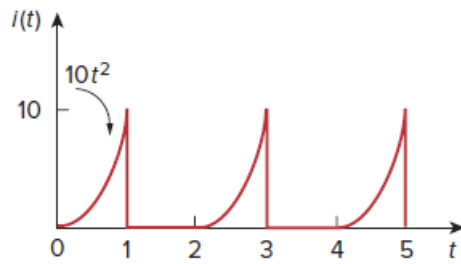


Problem

Obtain the rms value of the current waveform shown in Fig. 11.63.

Figure 11.63:



Step-by-step solution

Step 1 of 1

$$I_{rms}^2 = \frac{1}{2} \left[\int_0^1 (10t^2)^2 dt + \int_1^2 0 dt \right]$$

$$I_{rms}^2 = 50 \int_0^1 t^4 dt$$

$$I_{rms}^2 = 50 \times \frac{t^5}{5} \Big|_0^1$$

$$I_{rms}^2 = 10$$

$$\therefore \boxed{I_{rms} = 3.162 \text{ A}}$$